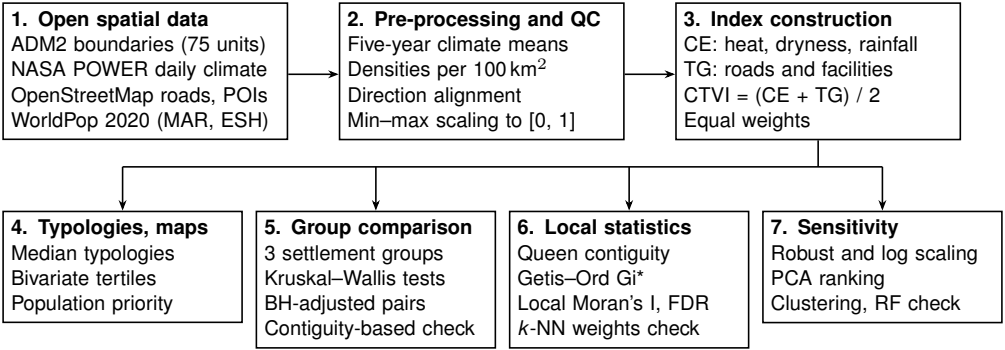
1. Open spatial data
ADM2 boundaries (75 units)
NASA POWER daily climate
OpenStreetMap roads, POIs
WorldPop 2020 (MAR, ESH)
2. Pre-processing and QC
Five-year climate means
Densities per 100 km2
Direction alignment
Min–max scaling to [0, 1]
3. Index construction
CE: heat, dryness, rainfall
TG: roads and facilities
CTVI = (CE + TG) / 2
Equal weights
4. Typologies, maps
Median typologies
Bivariate tertiles
Population priority
5. Group comparison
3 settlement groups
Kruskal–Wallis tests
BH-adjusted pairs
Contiguity-based check
6. Local statistics
Queen contiguity
Getis–Ord Gi*
Local Moran's I, FDR
k-NN weights check
7. Sensitivity
Robust and log scaling
PCA ranking
Clustering, RF check